AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX

MODEL EXAMINATION PAPER 2026 AND ONWARDS

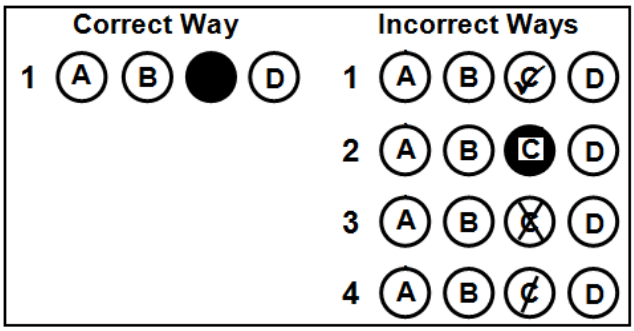
Chemistry Paper I

Time: 1 hour 10 minutes Marks: 40

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Model Paper 2026
For Teaching and Learning Only

INSTRUCTIONS:

1. Check your name and school information on the question paper and Optical Mark Recognition (OMR) sheet. Sign on both if the information is correct.
2. Read each question carefully.
3. Answer all questions on the provided OMR sheet within the allotted time. DO NOT mark your answers on the question paper.
4. There are 100 answer numbers on the OMR sheet. Use answer numbers 1 to 40 only.
5. Each question has four choices: A, B, C, and D. Choose ONE correct answer only. On the OMR sheet, completely fill in the circle corresponding to your choice with a pencil as shown in the example below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with an eraser, before filling in the new circle. Ensure your marks are dark and clear. Do not make stray marks or leave incomplete fillings as the OMR machine ONLY records what is in the circles.
7. Do not fold, tear, or damage the OMR sheet. Damaged sheets may be scanned incorrectly.
8. You may use a simple calculator if you wish.

1. The valency of iron (Fe) in $Fe(NO_3)_3$ is

   A. 0
   B. + 1
   C. + 2
   D. + 3

2. For a research project, a student investigates the presence of minerals, including their occurrence, chemical composition and changes over time.

   This study relates to the field of

   A. biochemistry.
   B. geochemistry.
   C. organic chemistry.
   D. polymer chemistry.

3. $H_3C^{\bullet}$ is a free radical because it

   A. is a stable unit.
   B. carries a net charge.
   C. possesses odd number of electrons.
   D. forms in the absence of light energy.

4. Consider the given features.

   - Purest form of substance
   - Possesses only one type of atom
   - Exists in bonded or non-bonded forms

   The substance that exhibits all these features is a/ an

   A. radical.
   B. mixture.
   C. element.
   D. compound.

5. Analytical chemistry is the branch of chemistry that deals with the techniques required for

   A. naming different substances.
   B. separating and identifying substances.
   C. reacting organic and inorganic compounds.
   D. changing the physical structure of compounds.

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6. A soft drink contains 0.5 moles of carbon dioxide (CO_2) dissolved in it. How many CO_2 molecules are present in the drink?

(**Note**: ${}^{12}_{6}C$ and ${}^{16}_{8}O$)

A. 3.01×10^{23} molecules
B. 6.02×10^{23} molecules
C. 1.50×10^{24} molecules
D. 3.01×10^{24} molecules

7. $\underline{\mathbf{w}}\,NH_{3(g)} + \underline{\mathbf{x}}\,O_{2(g)} \rightarrow \underline{\mathbf{y}}\,NO_{(g)} + 6H_2O_{(g)}$

To balance the given chemical equation, the values of **w**, **x** and **y** would be

	w	**x**	**y**
A	2	5	2
B	2	3	2
C	4	5	4
D	4	3	4

8. A compound **X** has an empirical formula C_4H_9 and its molecular mass is 114 g/mol.

Based on the given information, the molecular formula of compound **X** will be

(**Note**: ${}^{12}_{6}C$ and ${}^{1}_{1}H$)

A. C_6H_{11}
B. C_7H_{12}
C. C_8H_{18}
D. $C_{12}H_{27}$

9. A student adds a piece of zinc metal to hydrochloric acid in a test tube and records the following observations:

I. The zinc becomes smaller in size.
II. The solution remains colourless.
III. A gas is released with bubbling.
IV. The reaction vessel feels warm.

Which combination of observations definitely proves a chemical reaction has occurred?

A. I and II
B. I and III
C. II and IV
D. III and IV

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10. The percentage by mass of sulphur (S) in one molecule of sodium thiosulphate ($Na_2S_2O_3$) is

(**Note**: $^{23}_{11}Na$, $^{16}_{8}O$ and $^{32}_{16}S$)

A. 32.85%.
B. 40.51%.
C. 70.01%.
D. 91.43%.

11. In a molecule of ethyne (C_2H_2), how many electron pairs are involved in the bonding between carbon-carbon atoms?

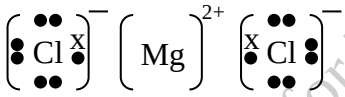
A. 2
B. 3
C. 4
D. 5

12. Read the given characteristics.

- Usually dissolves in water
- Carries partial charges on atoms
- Uneven distribution of electrons

Based on the given characteristics, an example of polar covalent molecule is

A. O_2
B. Cl_2
C. H_2S
D. NaCl

13. 

(2,8,8) (2,8) (2,8,8)

The given dot and cross structure exemplifies the formation of

A. ionic bonds.
B. metallic bonds.
C. covalent bonds.
D. coordinate covalent bonds.

14. Metals are good conductors of electricity due to the presence of

A. freely moving electrons.
B. number of positive charges on the ions.
C. atoms in the form of rows one above the other.
D. electrostatic forces of attraction between oppositely charged ions.

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15. Two atoms, **X** and **Y**, have electronegativity values of 1.0 and 3.5, respectively.

The type of bond that will form between **X** and **Y** is

A. ionic bond.
B. metallic bond.
C. polar covalent bond.
D. non-polar covalent bond.

16. The isotope that is CORRECTLY matched with its application is

	Isotope	**Application**
A	Uranium-235	used in nuclear reactors for power generation
B	Iodine-131	used in industries as a fuel source
C	Technetium-99	used in determining the age of fossils
D	Sodium-24	used in nuclear weapons for energy release

17. Which of the following options shows the sub-atomic particles of a species having a +1 charge on it?

	Number of Electrons	**Number of Protons**	**Number of Neutrons**
A	18	20	20
B	18	19	20
C	10	9	10
D	10	8	8

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18. Rutherford's atomic model explains all of the following statements EXCEPT that the

A. volume occupied by the nucleus is very small.
B. positively charged particles exist inside the nucleus.
C. entire mass of the atom is concentrated in the nucleus.
D. energy emitted by the electrons forms a line spectrum.

19. The electronic configuration of potassium $\left({}^{39}_{19}\mathrm{K}\right)$ is

A. $1s^2, 2s^2, 2p^6, 3s^2, 3p^5$
B. $1s^2, 2s^2, 2p^6, 3s^2, 3p^6$
C. $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$
D. $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2$

20. How many sub-shells are present in an L-shell?

A. 1
B. 2
C. 3
D. 4

21. A transition metal has the following characteristics:

- Weak metallic bonds
- Low melting point

Based on these characteristics, the metal is identified as

A. iron.
B. copper.
C. mercury.
D. titanium.

22. In the periodic table, an element with atomic number 14 is placed in period

A. 1
B. 2
C. 3
D. 4

23. Which of the following statements is FALSE regarding ionisation energy of elements?

A. The first ionisation energy of lithium is less than that of beryllium.
B. The second ionisation energy of magnesium is less than that of sodium.
C. It is the amount of energy required to remove an electron from a gaseous atom or ion.
D. It is the tendency of an atom or molecule to attract electrons of other atom or molecule to itself.

24. If a solid substance conducts electricity, then it must be a/ an

A. metal.
B. non-metal.
C. ionic compound.
D. covalent compound.

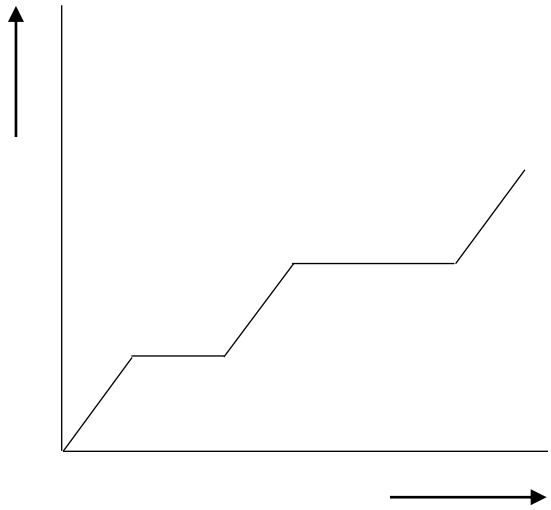
25. Barium belongs to group IIA while chlorine belongs to group VIIA of the modern periodic table.

If barium reacts with chlorine, then their product is

A. BaCl
B. $BaCl_2$
C. Ba_2Cl
D. Ba_2Cl_2

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26. The heating curve below shows the temperature of a solid substance as it is heated over time. At points **X** to **Y**, the temperature remains constant even though heat is continuously added.



Aligning with the kinetic theory of matter, this occurs because the added energy

A. stops the transfer of heat to the substance.
B. increases the kinetic energy of particles.
C. overcomes intermolecular forces.
D. disables particles' movement.

27. At constant temperature and pressure, which of the following gas molecules will diffuse the fastest through air?

(**Note**: Atomic mass of O = 16 amu, Cl = 35.5 amu, N = 14 amu, C = 12 amu and H = 1 amu)

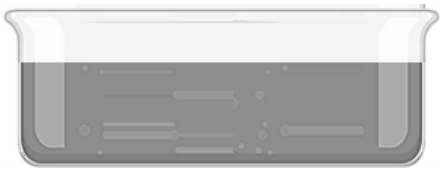
A. $O_{2(g)}$
B. $Cl_{2(g)}$
C. $NO_{2(g)}$
D. $CH_{4(g)}$

28. At constant temperature, if the volume of a gas in a container is reduced to half, then the pressure on the gas will

A. be doubled.
B. reduce to half.
C. remain the same.
D. reduce to a quarter.

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29. In the given diagrams, the rate of evaporation of water at room temperature, i.e., 25°C would be the greatest in

	
A	B
	
C	D

30. Ahmed plotted a graph representing Charles's Law. He placed temperature on ***x*-axis** and volume of one mole of an ideal gas on ***y*-axis**.

At a constant pressure, Ahmed obtained a straight line which intersects the ***x*-axis** at

A. – 273.16°C.
B. – 273.16 K.
C. 0°C.
D. 300 K.

31. What is the molarity of a solution containing 32 g of potassium sulphate (K_2SO_4) in 350 cm^3 of solution?

(**Note**: Molar mass of K_2SO_4 = 174 g mol^{-1})

A. 0.06 M
B. 0.09 M
C. 0.52 M
D. 64.37 M

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32. The given diagram shows an area covered in fog.



Fog is a solution of

A. gas in liquid.
B. liquid in gas.
C. solid in liquid.
D. liquid in liquid.

33. An INCORRECT difference between solution and suspension is

	Solution	Suspension
A	it passes through the filter paper.	it leaves residue on filtration.
B	it forms homogeneous mixture.	it forms heterogeneous mixture.
C	it does not show Tyndall effect.	it may show Tyndall effect.
D	it appears translucent.	it appears transparent.

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34. A teacher asks a student to dissolve 2.5 g of sodium carbonate in 50 g of water. What will be the concentration of the prepared solution?

A. 0.048%
B. 0.050%
C. 4.762%
D. 5.000%

35. In a science laboratory, Mariam prepared a mixture of SiO_2 and $K_2SO_4.Al_2(SO_4)_3.24H_2O$ in water. She then applied the process of filtration on the mixture.

An inference Mariam must have made through this experiment is that

A. SiO_2 is insoluble in water.
B. SiO_2 easily passes through the filter paper.
C. $K_2SO_4.Al_2(SO_4)_3.24H_2O$ is an amorphous solid.
D. $K_2SO_4.Al_2(SO_4)_3.24H_2O$ easily settles down in water.

36. Which of the following CORRECTLY matches the redox reaction with its inference?

	Redox Reaction	Inference
A	$2HI_{(g)} + Cl_{2(g)} \rightarrow I_{2(g)} + 2HCl_{(g)}$	Hydrogen iodide is reduced to iodine.
B	$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$	Magnesium is reduced to magnesium chloride.
C	$CuSO_{4(aq)} + Mg_{(s)} \rightarrow MgSO_{4(aq)} + Cu_{(s)}$	Copper sulphate is reduced to copper.
D	$H_2S_{(g)} + Cl_{2(g)} \rightarrow S_{(s)} + 2HCl_{(g)}$	Hydrogen sulphide is reduced to sulphur.

37. The gas(es) produced during the electrolysis of fused sodium chloride in Down's cell is/ are

I. $Cl_{2(g)}$
II. $O_{2(g)}$
III. $H_{2(g)}$

A. I only.
B. III only.
C. I and II.
D. I and III.

38. When Altamash places iron nails in a copper sulphate solution, the solution's colour changes from blue to light green.

What does this observation indicate about the reaction?

A. The melting of iron nails in the solution
B. The formation of iron sulphate in the solution
C. The dissolution of metallic copper in the solution
D. The corrosion of iron nails in copper sulphate solution

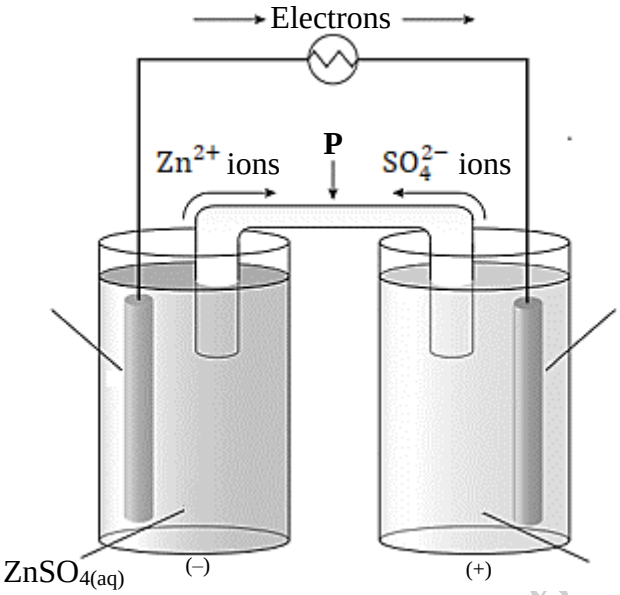
39. A solution of 1.0 M sodium chloride conducts electricity well, while a 1.0 M sugar solution does not.

Based on the difference in electrical conductivities of the two given solutions, it can be inferred that

A. sodium chloride absorbs more water, increasing its conductivity.
B. sugar has larger molecules, making it harder to conduct electricity.
C. sodium chloride dissociates into ions, but sugar remains as discrete molecules.
D. both solutions have same concentrations, but sugar has a higher molecular mass.

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40. Consider the given galvanic cell.



The component **P** is identified as a/ an

A. battery.
B. ammeter.
C. voltmeter.
D. salt bridge.

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END OF PAPER

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